MINING DEALER BEST PRACTICE

A Method for Chassis Center Tube Alignment on 793D Mining Trucks

Maintenance and Repair | Application | Component Life Management | **Component Renewal (CRC)** | MARC Management

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1.0 Introduction

Structural repairs on large mining equipment are critical activities, due to the large size of the parts and the need to release the equipment with high reliability and Low Repair Time.

One of the critical items, when thinking about **793D** truck fleets, is the chassis repairs, which present several points of cracks, aggravated mainly by the severity of the operation, and overload, besides the low reliability of Maintenance in items such as suspension and initial crack repairs.

In the **793D**, the region in the Center Tube usually has cracks that can be repaired, in some situations total replacement of the part is required.

The repair instruction "**M0086585:** Procedure to Repair the Frame on Certain 793 through **793D** Off-Highway Trucks" shows the replacement of the Center Tube, performing the alignment using the chassis as reference, however, due to the removal of the old Center Tube, the alignment needs attention to verify the measurements, and the need to perform this adjustment with the part lifted.

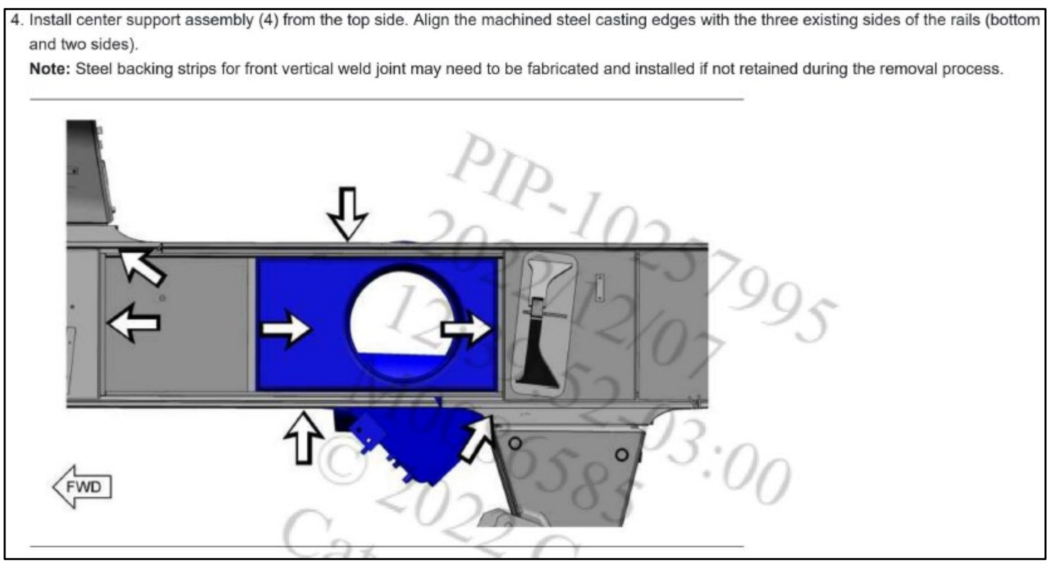

Image 01. Alignment of the Center Tube according to procedure M0086585

To improve the alignment of the Center Tube, this Best Practice presents tooling designed to perform the activity, making the alignment more precise and reducing the effort to perform the activity.

2.0 Best Practice Description

To improve the alignment of the Center Support during replacement, a tooling that uses the equipment's central pin clamping bearings has been developed, see the images below:

Image 02. Tooling Support made before the removal of the Center Tube (open image)

Image 03. Tooling Support made before the removal of the Center Tube (More focused image)

The Tooling Support is made with the central pin fixing bearings **(PN 8X-2944),** which must be fixed on the Center Tube by bolt **9X-6151** and then the fixing structure is made, (picture example) but another profile can be used. The Center Support must be made before removing the damaged tube and fixed to the chassis, so it is possible to obtain the installation reference for the new tube. The sequence of execution of the central pipe replacement using the tooling and the sketch with reference dimensions is presented below.

3.0 Implementation Steps

The Tooling Support is made with the Center Tube that will be removed. The bearings are fixed on the Center Tube and then a support is made so that the bearings are fixed in relation to the chassis. The image below shows the bearing and support:

Image 04. Making the Tooling Support before removing the Center Tube (more focused image)

After the tool is installed, the bearing retaining bolts are removed and the process of removing the damaged Center Tube is started. The picture below shows the removal of the Center Tube.

Image 05: Removal of the damaged Center Tube

The images below show the damaged Center Support removed and the template installed, thus maintaining the positioning reference for the new Center Tube to be installed.

Image 06. Tooling positioned after removal of the Center Tube

Image 07: Tool positioned during preparation for installation of the new Center Tube

After preparation for the new part, installation is made using the Tooling Support as reference. The images below show the installation and alignment of the new Center Support.

Image 08: Alignment of the Center Tube in the Chassis

Image 09. New Center Tube fixed to the Chassis

After alignment and fixing the Tooling Support on the new part, the welding process of the Center Tube on the chassis is started, and after this process the removal of the tooling is performed.

4.0 Benefits

The main benefits related to this improvement are related is:

- Less labor effort to perform the alignment.
- Less exposure of the technicians during the installation process.
- Better part alignment.

5.0 Resources Required

The main items and **PN** has shown below:

Qty	PN	Description
02	**8X-2944**	Bearing
08	**9X-6151**	Bolt
6,0 m	-	Metalon ¼ (Dimensions: 80 x 80) cm

The dimensions for the tie rods may vary. Their main function is to keep the bearings fixed to the chassis. The dimensions provided meet the technical design, however field adjustments may be necessary and should be considered during tooling manufacturing and use.

6.0 Supporting Attachments / References

M0086585: Procedure to Repair the Frame on Certain **793** through **793D** Off-Highway Trucks

7.0 Related Best Practices

N/A

8.0 Acknowledgements

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